MIL-R-43399A

18 August 1970

SUPERSEDING
MIL-R-43399
7 January 1966

MILITARY SPECIFICATION

RACK, MOBILE (FOR ROLL-THRU REFRIGERATORS AND ROLL-THRU FOOD WARMERS)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of metal, mobile, rack used to hold bun pans and steam table pans (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-C-77 - Casters, Institutional Duty
QQ-A-200/9 - Aluminum Alloy Bar, Rod, Shapes, and Tube, Extruded, 6063.

MILITARY

MIL-B-2316 - Bakery Equipment, Including Units, Assemblies, Repair Parts and Tools, Preparation for Delivery of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, and figures required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 7320

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bid or request for proposal shall apply.

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 2 - Food Service Equipment
Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, Box 1468, Ann Arbor, Michigan 48106.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

* 3.1 First article.- This specification contains provisions for first article test and approval (see 4.2, 6.2, and 6.3).

3.2 Standard product.- The racks delivered under this specification shall be the manufacturer's standard commercial product except for any changes necessary to meet specification requirements. Component parts of the racks need not be the product of the same manufacturer.

* 3.3 NSF standard.- Racks shall conform to the applicable requirements of NSF Standard No. 2.

* 3.3.1 NSF certification.- The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the shelving he proposes to furnish under this specification meets the applicable requirement of NSF Standard No. 2 as follows:

(1) Listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished shelving shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, or

(2) A certification for the racks issued by NSF under their special one time evaluation/certification service shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, or

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(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the racks are being procured, indicating that the racks have been tested and conform to NSF Standard No. 2 shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to commencing production.

* 3.4 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for racks provided the completed item complies with all provisions of this specification.

3.4.1 Aluminum alloy.- Aluminum alloy shall conform to 6063-T6 of QQ-A-200/9.

* 3.5 Design and construction.- The rack shall be constructed as specified herein and as illustrated in figure 1. Figure 1, except for wraparound bumpers, overall dimensions, spacing of pan runners, capacity of pans, and details of caster construction, is for illustrative purposes only and is not intended to preclude other racks otherwise meeting the requirements of this specification. All parts of the rack, excluding casters, shall be fabricated of the material specified in 3.4.1. The base of the frame shall be turned down to provide a mounting area for securing a wraparound rubber bumper. The rubber bumper shall extend out beyond the bottom edges of the rack not more than 1 inch. In addition two 1/2-inch-diameter tie rods shall be placed as shown on figure 1. Each rack shall be provided with four casters conforming to type II, class B, 5 inch diameter, style c, mounting optional, of FF-C-77. Two casters diagonally opposite each other shall be equipped with brakes. The rack shall accommodate a minimum of eighteen pair of tray slides fabricated as shown on figure 1. The rack, including casters, shall show no evidence of permanent deformation, loss of rigidity, or other evidence of failure, when tested as specified in 4.4.1. The casters shall roll and swivel freely without binding and shall show no evidence of failure when tested as specified in 4.4.2.

3.6 Finish.- Aluminum alloy finish shall be the manufacturers standard finish provided the finish meets the NSF requirements.

* 3.7 Marking for identification.- Each rack shall be permanently and legibly marked with the manufacturer's name or trademark readily identifiable with said, manufacturer.

3.8 Workmanship.- The racks shall not be dirty or streaky. Welds shall be adherent and complete. Scale, flux, pits, slag, or splatter shall be removed. Rough or sharp edges, burrs, and slivers shall be removed.

4. QUALITY ASSURANCE PROVISIONS

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* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects listed in table I, for dimensions specified herein, and tested in accordance with 4.4.1 and 4.4.2. The presence of any defects listed in table I, any dimensions not within specified requirements, or failure to pass any of the tests in 4.4.1 and 4.4.2 shall be cause for rejection of the preproduction unit.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.3.2 End item inspection.- The inspection lot shall consist of all racks offered for inspection at one time. The sample unit shall be one completely assembled rack.

4.3.2.1 Visual examination.- Examination shall be made of the racks for defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I.- Classification of visual defects

Examine	Defect	Classification Major	Minor
Finish	Not manufacturers standard finish		X
	Rough or sharp edges, burrs, or slivers	X	
	Dirty or streaky		X
Design, construction and workmanship	Component missing	X	
	Component loose		X
	Not material specified	X	
	Rubber bumper missing or does not extend beyond the bottom edges	X	
	Tie, rods missing or not placed as shown on figure	X	
Welding	Welding not adherent or complete	X	
	Scale, flux, pits, slag, or splatter not removed		X
Marking for identification	Missing, incomplete, illegible		X

4.3.2.2 Dimensional examination.- Examination shall be made of the completed racks to determine compliance with this specification. Any dimensions not within specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

* 4.3.2.3 Testing of the end product.- The racks shall be tested as specified in 4.4.1 and 4.4.2. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.3 Standards compliance.- Proof of compliance with the requirements of 3.3.1 shall be submitted to the Government representative.

4.3.4 Examination of preparation for delivery.- Examination shall be made for preservation, packaging, packing, and marking in accordance with MIL-B-2316.

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4.4 Tests.-

* 4.4.1 Load test.- The rack shall be filled to capacity with 18-inch by 26-inch bun pans and each pair of tray slides shall support a uniformly distributed load of 20 pounds, including pan weight, for a minimum total load of 400 pounds over the entire rack assembly. The rack shall be rolled a distance of 3 feet in all directions to determine compliance with the rigidity requirement of 3.5. Failure of the rack to remain rigid, any deformation, or loosening of the slides shall constitute failure of this test.

* 4.4.2 Caster Test.- The rack, loaded with 400 pounds, shall be rolled up and over an obstacle 10 times at 2 miles per hour. The obstacle shall have vertical faces a minimum of 1/4 inch high, shall be at least 12 inches wide, and shall be long enough to allow both leading casters to strike the obstacle at the same time. Upon completion, failure of the casters to meet the requirements of 3.5 shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking.- Preservation, packaging, packing, and marking for level A, B, or C shall be in accordance with MIL-B-2316 (see 6.2).

6. NOTES

6.1 Intended use.- The rack covered by this specification is intended for use in storing foodstuffs in refrigerator - warmer cabinets conforming to MIL-C-43427. The pan sizes, in inches, may be stored as follows:

(a) Eighteen bun pans - 18 wide by 26 long by 1 deep.
(b) Thirty-six steam table pans - 12 wide by 20 long by 2-1/2 deep.
(c) Eight steam table pans - 12 wide by 20 long by 4-1/2 deep and fifteen bun or thirty 2-1/2 deep steam table pans in "a" and "b".
(d) Any combination of pans in "a" and "b" or eight pans in "c" and a combination of pans in "a" and "b".

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) First article (see 3.1, 4.2, and 6.3).
(c) Selection of applicable levels of packaging and packing (see 5.1).

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6.3 First article.- The item covered by this specification requires first article test and approval under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 7320-0265

Review activities:

Army - MD
Navy - SA

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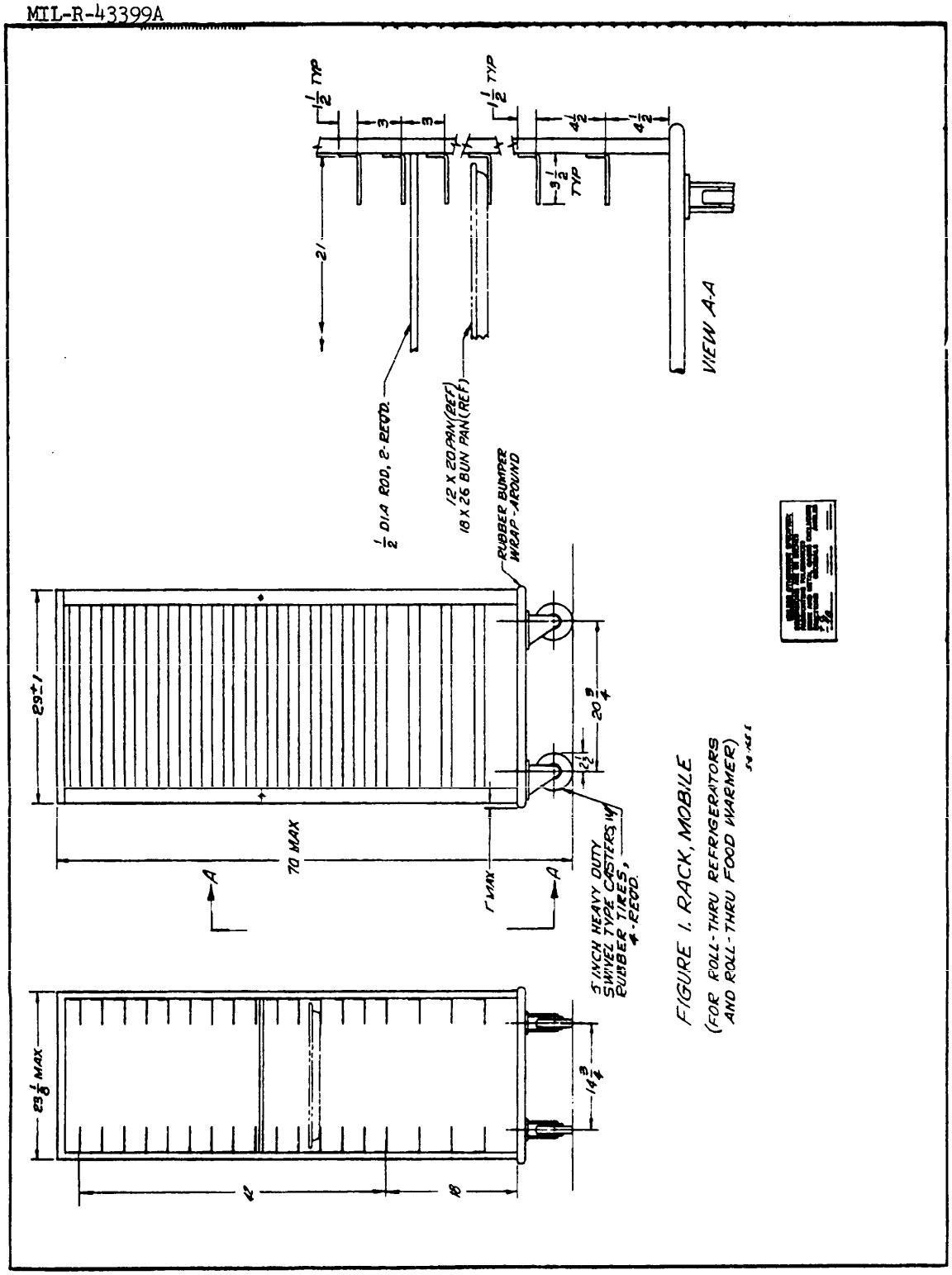


FIGURE 1. RACK, MOBILE

(FOR ROLL-THRU REFRIGERATORS AND ROLL-THRU FOOD WARMER)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Rack, Mobile (For Roll-Thru Refrigerators And Roll-Thru Food Warmers)MIL-R-43399A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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